

LinkedIn Profile — Copy/Paste Descriptions

Headline

Doctoral Researcher (AI-Driven Maintenance for Solar Minigrids), Birmingham City University | Business Intelligence Manager, ECAM Engineering | Mechatronics Engineer

About

Mechatronics engineer working across manufacturing, data, and applied AI.

As Business Intelligence Manager at ECAM Engineering, I lead data optimisation from shop-floor operations to strategic planning — improving operational efficiency, surfacing growth opportunities, and supporting long-term competitiveness.

I am also a Doctoral Researcher at Birmingham City University, researching AI-driven maintenance optimisation for solar minigrids under the UKRI EPSRC Ayrton Challenge programme — predictive maintenance, digital twins, and energy access in off-grid contexts.

Previously, as a Knowledge Transfer Partnership Associate, I led a manufacturing digitalisation project and built proprietary business software bridging academic research and industry practice.

Core areas: Industry 5.0, cyber-physical systems, data transformation, machine learning, and manufacturing analytics.

Experience

Doctoral Researcher — Birmingham City University

February 2026 – Present

- Researching AI-driven maintenance optimisation for solar minigrids under the UKRI EPSRC Ayrton Challenge programme.
- Focus areas: predictive maintenance, digital twins, and energy access in off-grid contexts.

Business Intelligence Manager — ECAM Engineering Limited

August 2025 – Present

- Lead data strategy from shop-floor operations to executive planning.
- Track production data, capacity, OEE, and KPIs to target operational improvements.
- Build dashboards and reports that inform decision-making across the business.
- Developed and maintain a lightweight MRP-II system for multi-level BOM production scheduling and forecasting.
- Monitor market and competitor trends; advise leadership on growth gaps and AI-led efficiency.

Business Systems and Technology Analyst — ECAM Engineering Limited

July 2023 – July 2025

- Led a digitalisation project improving visibility and traceability across the manufacturing process.
- Embedded academic research from Birmingham City University into ECAM's operations.
- Aligned digital transformation initiatives with business goals to deliver measurable impact.

Knowledge Transfer Partnership Associate — Birmingham City University

July 2023 – July 2025

- Bridged Birmingham City University and ECAM Engineering to drive innovation and growth.
- Integrated academic insight into operations, improving processes and efficiency.
- Fed real-world manufacturing challenges back into the university's research base.

Graduate Project Engineer (DigitME2) — University of Central Lancashire

February 2022 – June 2023

- Developed DigitME2 web (PHP, MySQL) and Android (Java) applications for SMEs.
- Built MES time-tracking, inventory, and OEE monitoring tools.
- Automated testing with Python and Selenium.
- Built ARIMA models for time-series prediction on simulated datasets.

Research Intern — Autonomous Vehicles (Erasmus+) — Arab Academy for Science, Technology & Maritime Transport

June 2021 – September 2021 · Grade: Distinction · European Commission funded

- Researched low-cost object detection (ultrasonic, infrared, camera) for autonomous navigation.
- Integrated Arduino and Raspberry Pi with MPU-6050 sensors for lane-following and vehicle stability.
- Implemented MATLAB and Python algorithms for pothole detection and obstacle avoidance.

Assembly Operator — Baxi Heating

November 2021 – February 2022

- Completed final boiler assembly to quality and safety standards.
- Ran pressure-holding tests and resolved assembly-line issues with the production team.

Sortation Associate — Amazon Logistics

September 2020 – January 2021

- Sorted and dispatched parcels by delivery route in a high-volume environment.

Production & Warehouse Operator — Royal Sanders (UK) Limited

May 2020 – September 2020

- Operated personal-care production lines and supported warehouse supply to meet daily targets.

Customer Service Associate — Amazon Development Centre India

August 2018 – August 2019

- Resolved amazon.co.uk customer queries via chat and email (orders, Prime, refunds, deliveries).
- Handled high volumes during peak periods with strong resolution rates.